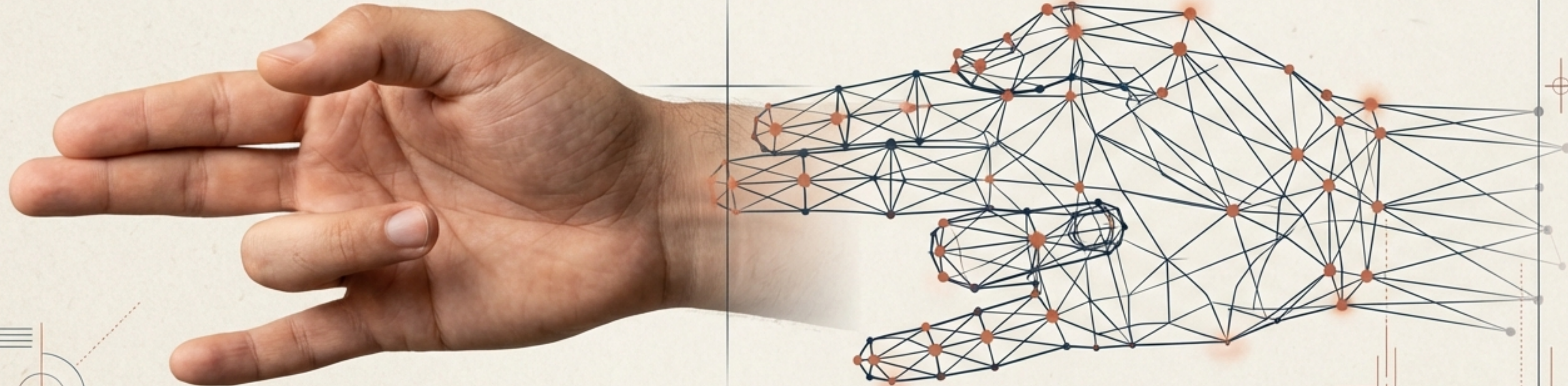


Arabic Hand Sign Language Recognition AI

Bridging the Communication Gap Through Computer Vision

مشروع التعرف على لغة الإشارة العربية باستخدام الذكاء الاصطناعي



The Silence of Millions

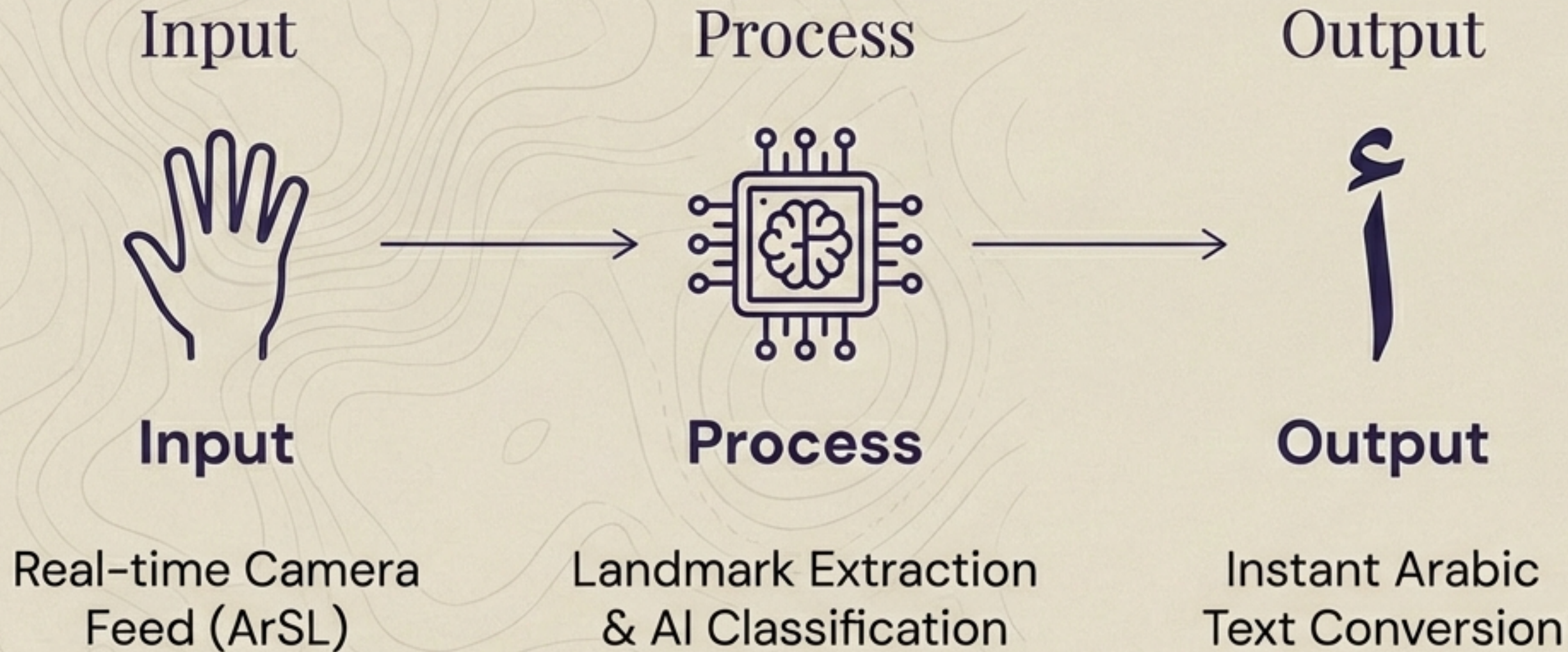
10 Million+

Over 10 million deaf and hard-of-hearing individuals in the Arab world rely on sign language.

A profound gap exists between them and the hearing community, often requiring human interpreters for basic interaction.

Mission: Real-time Translation for Independence and Inclusion

The Translation Bridge



Core Applications:

- Accessibility
- Education
- Biomedical Research

Technical Foundation



Vision: MediaPipe HandLandmarker



Logic: Scikit-learn MLP Classifier

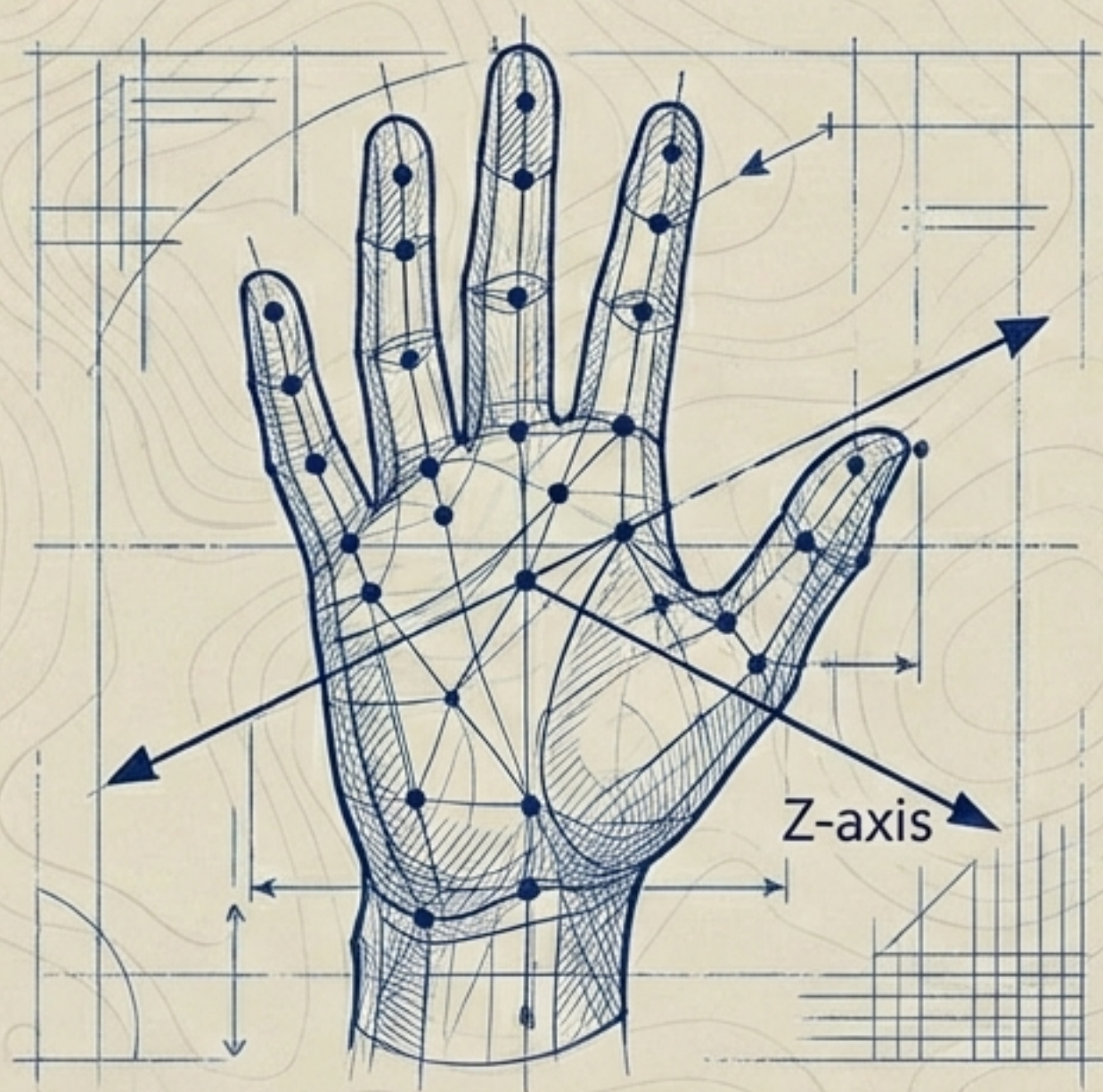


Core: Python Processing

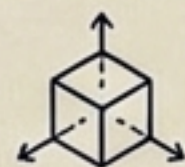


Interface: Streamlit Web App

Navigating Complexity

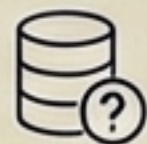


3D Spatial Logic (X, Y, Z) in DM Sans



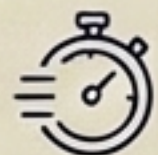
1. 3D Spatial Logic

Capturing depth is critical. The system processes 63 distinct features (21 landmarks $\times$ 3 coordinates) to understand gesture nuance.



2. Data Scarcity

Arabic Sign Language datasets are scarce and non-standardized compared to Western counterparts.

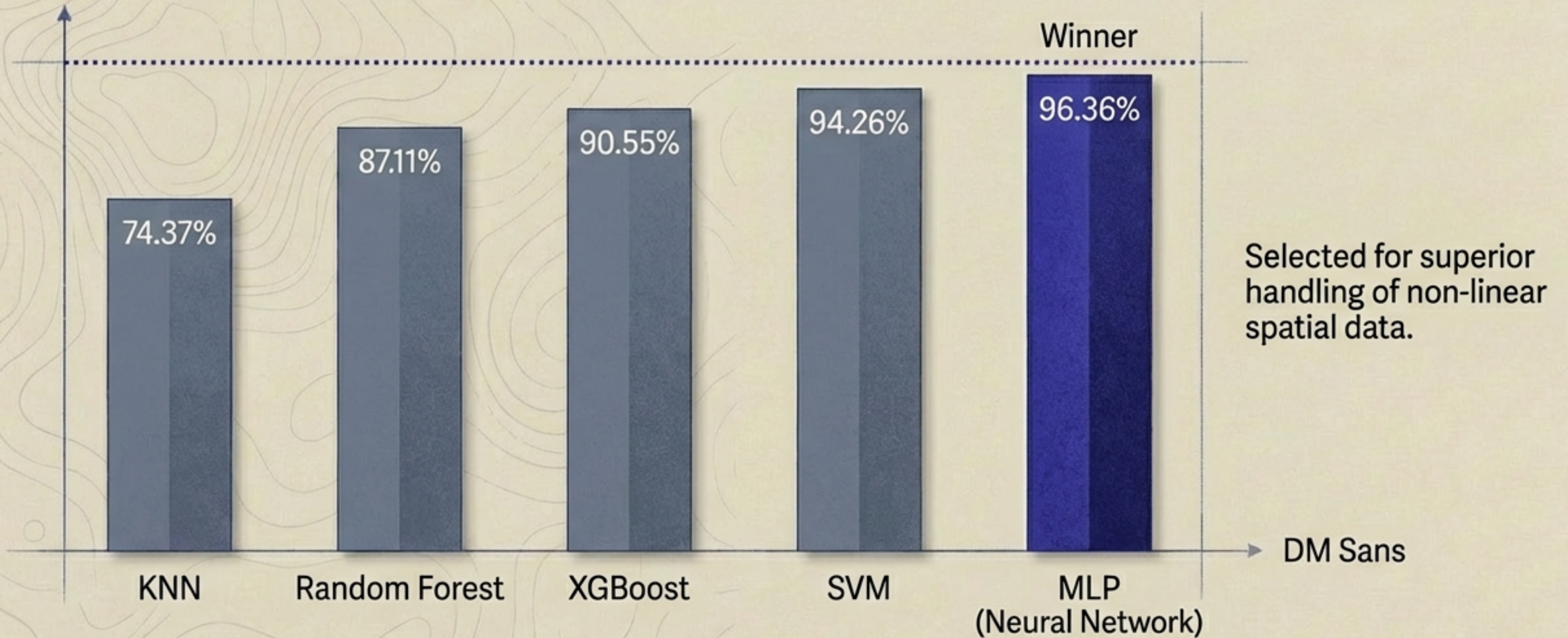


3. Real-Time Constraints

Balancing low-latency speed with high-accuracy detection.

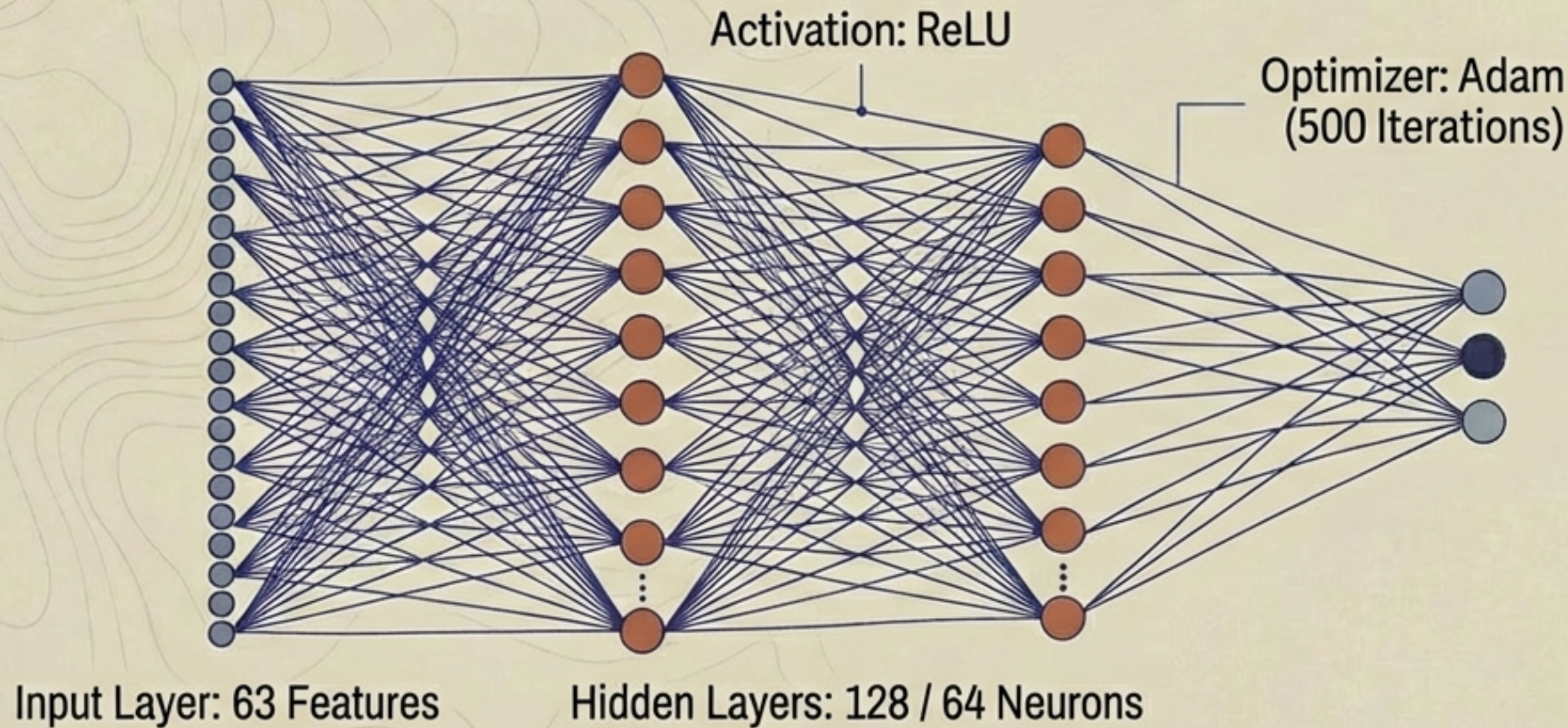
Selecting the Brain

Algorithmic Benchmarking Results



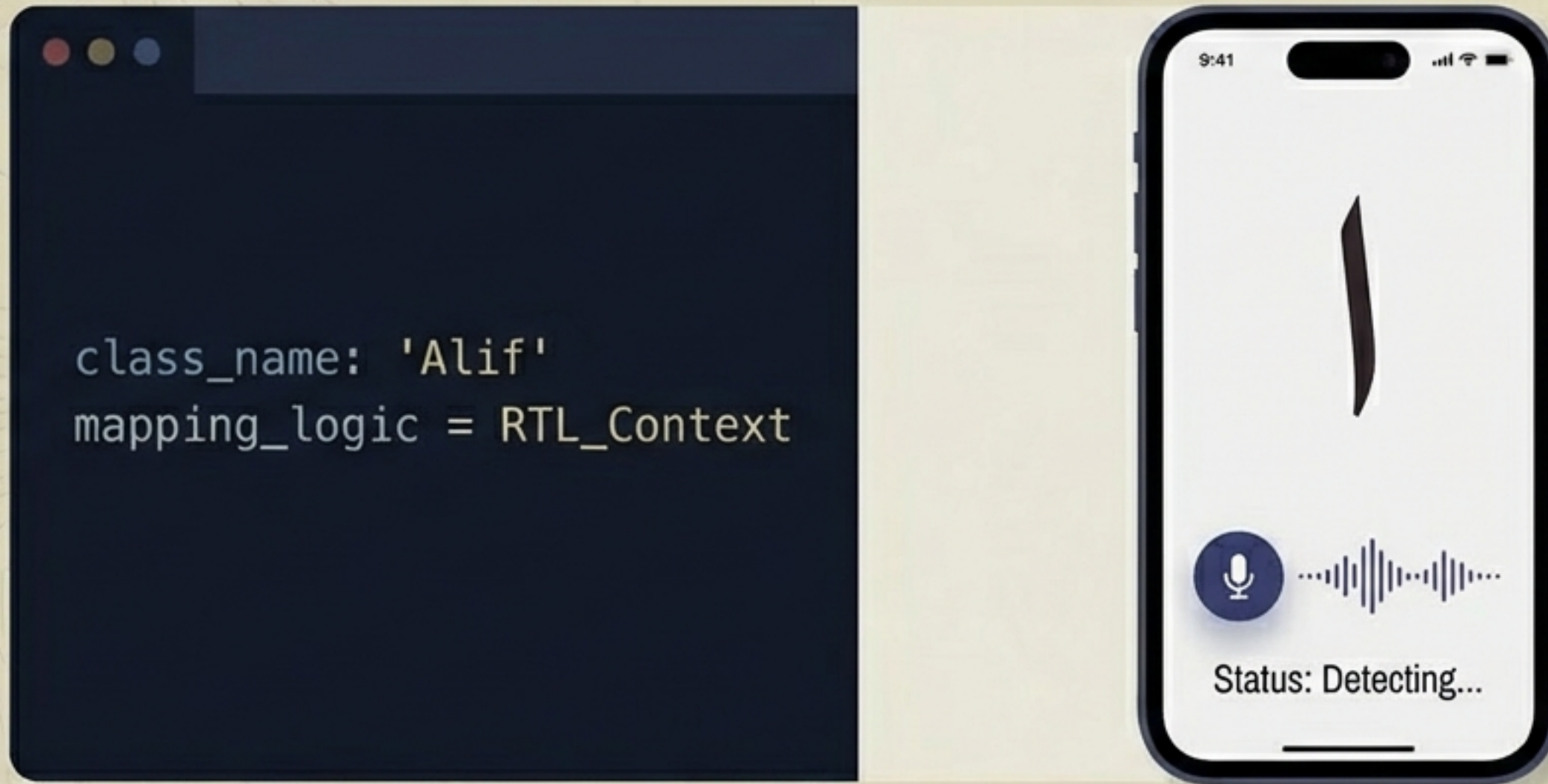
The Multi-Layer Perceptron (MLP)

96.36% Accuracy



Why it won: The Neural Network architecture proved superior at mapping the complex, non-linear relationships of 3D hand landmarks.

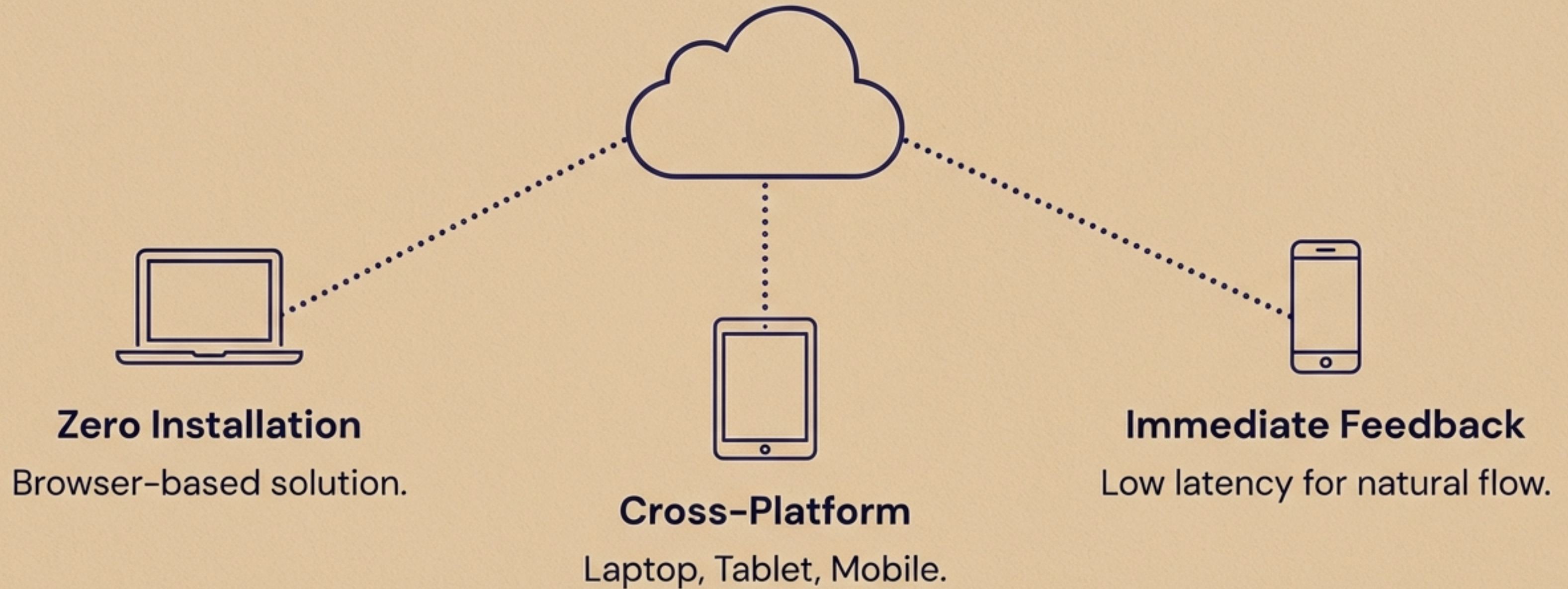
Designed for the Arabic Language



- RTL Integration: Seamless Right-to-Left interface design.
- Character Precision: Rendering 31 letters + special characters (e.g., ﻻ).
- Dialect Awareness: Built for regional signing variations.

Accessible Anywhere

Deployment via Streamlit Cloud



Beyond the Prototype

1. Sentence Logic

Moving from letter-by-letter to full natural conversation.

2. Two-Way Communication

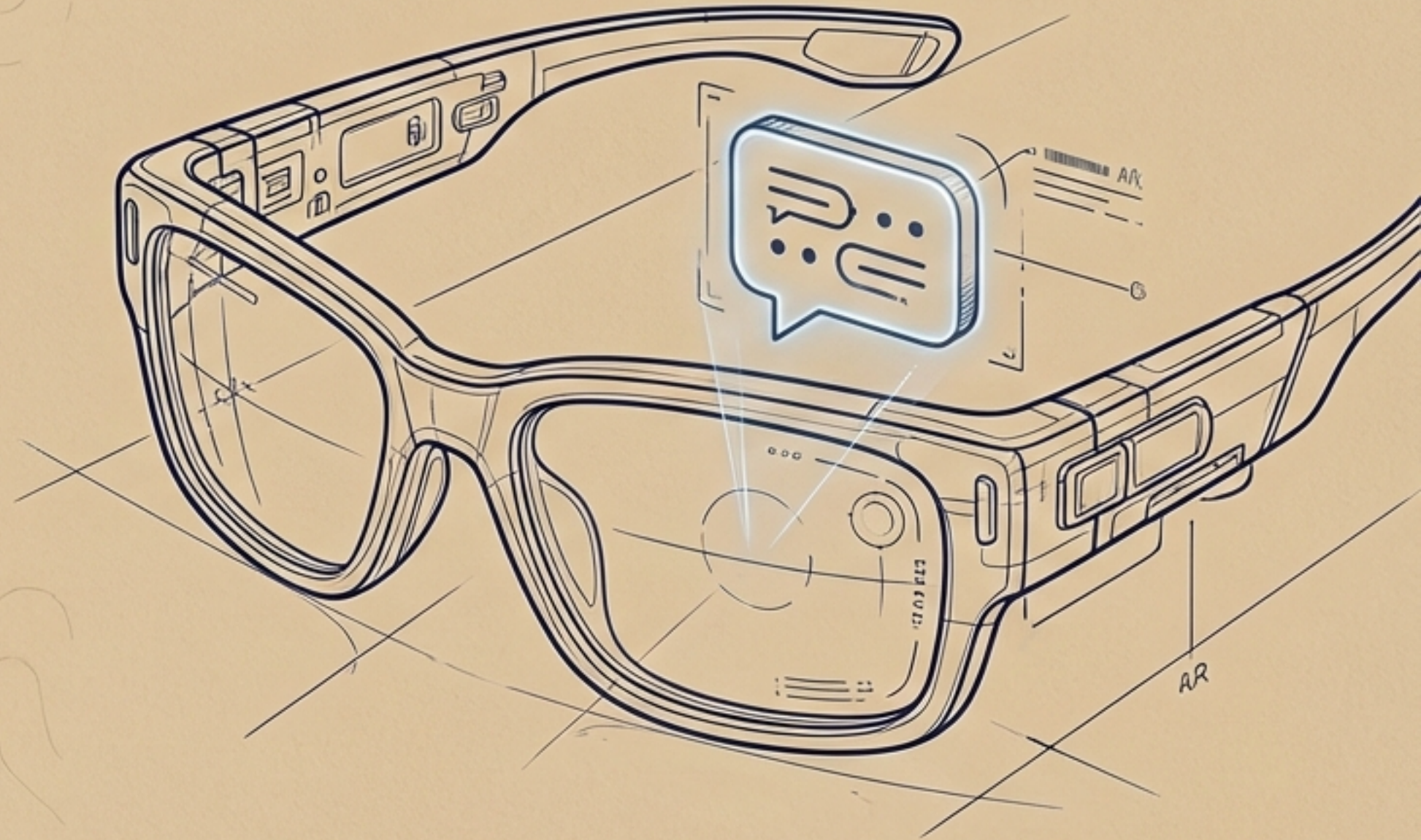
Text-to-Sign animations enabling hearing users to sign back.

3. Smart Integration

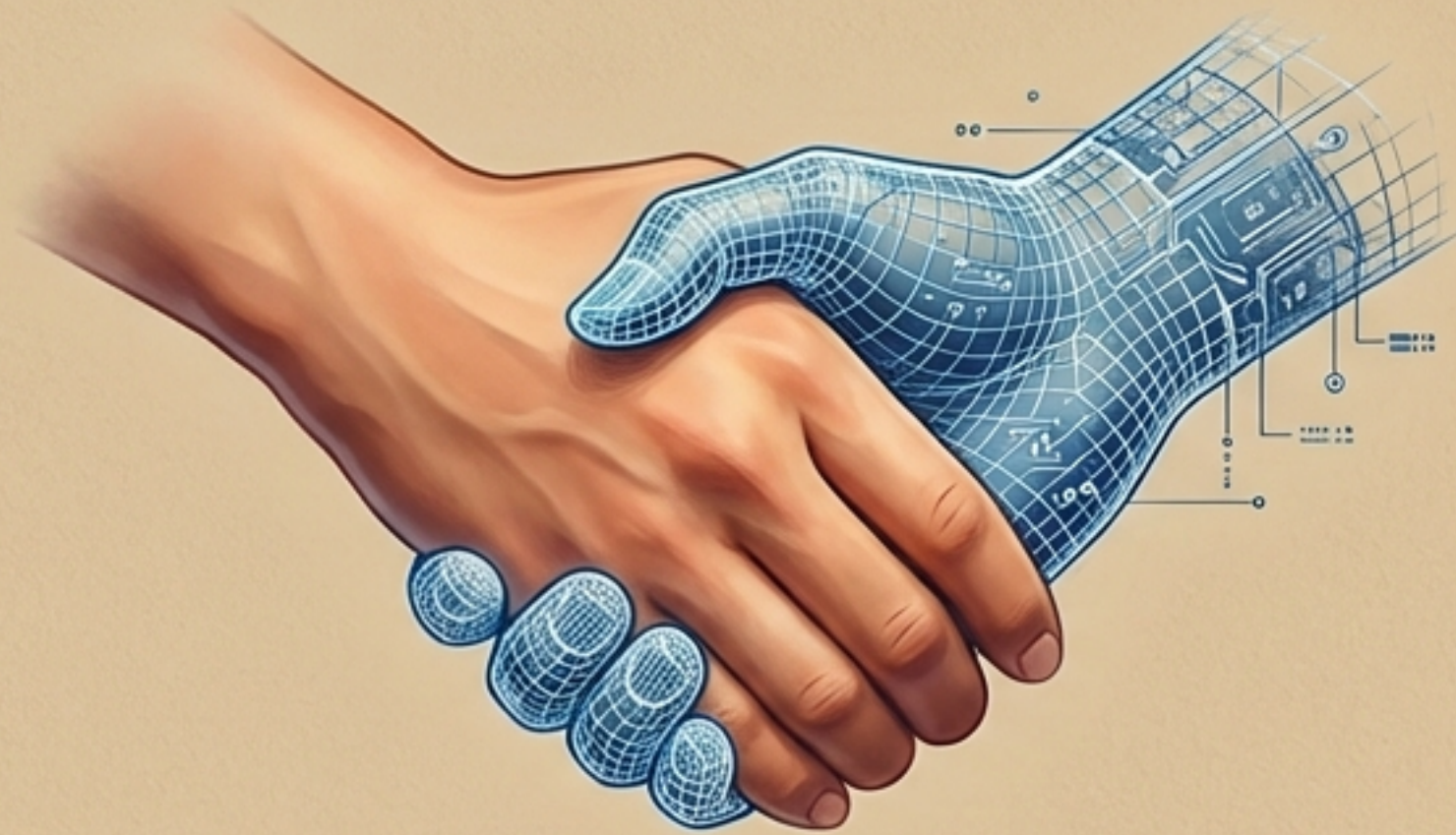
IoT wearables and Smart Glasses for hands-free assistance.

4. Universal Reach

Deployment in hospitals, schools, and government offices.



A World Without Barriers



"Technology is most powerful when it connects us.
Let's build a more accessible future."

Developed with ❤️ for the College of Artificial Intelligence
Department of Biomedical Applications

Q&A